

Biology Mocks 1 and 2 — mark scheme

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Marks in brackets are awarded line by line: a line of correct method or a correct marking point scores even when what follows it is wrong. Follow through an earlier wrong answer wherever the later reasoning is sound — the calculation questions in Section 3 are written so that this matters. The note under each question names the specific mistake that question was built to catch; across two papers those notes are meant to be read as a set, because the same three or four errors account for most of the marks lost.

Biology Mock 1 — mark scheme

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- 1 (a) the **diaphragm** [1] [2]
 (b) it **contracts** and flattens, moving down [1]
"The diaphragm goes down" without the word contracts scores 0 in (b). The muscle action is the mark; the movement is the consequence.
- 2 (a) about **16%** oxygen [1] [2]
 (b) about **4%** carbon dioxide [1]
Accept 15–17% and 3–5%. A candidate who writes 0.04% for expired carbon dioxide has memorised the inspired figure — worth pointing out, because the hundredfold difference is the whole basis of the limewater test.
- 3 (a) the **pulmonary vein** [1] [2]
 (b) the **left atrium** [1]
Pulmonary artery in (a) is the standard error. If they also wrote "right atrium" in (b) they have the whole right side of the heart mirrored — worth a two-minute redraw rather than a correction.
- 4 (a) the **tricuspid** valve (accept right atrioventricular valve) [1] [2]
 (b) it prevents blood flowing backwards into the atrium when the ventricle contracts [1]
"It stops the blood" is not enough for (b) — the direction and the moment both matter.
- 5 Any two of: **red blood cells** — carry oxygen as oxyhaemoglobin; **white blood cells** — defend against microbes by engulfing them or making antibodies; **platelets** — cause blood to clot [2] (1 for each component with a correct function) [2]
A component with no function, or a function with no component, scores nothing. Plasma is excluded by the question.
- 6 (a) **habitat**: the place where an organism lives [1] [2]
 (b) **community**: all the populations of all the different species living in an area [1]
The word "all" is doing the work in (b). A community of one species is a population, and that confusion runs through the whole unit.
- 7 (a) the **algae** [1] [2]
 (b) trophic level **3**, a **secondary consumer** [1]
Level 2 is the commonest wrong answer — counting the water flea as level 1 because it is the first animal. Count from the producer.
- 8 (a) **bacteria** and **fungi** [1] (both needed) [2]
 (b) **mineral ions** / nutrients such as nitrates [1]
"Nutrients" alone is acceptable; "energy" is not, and is the answer to watch for — it is the flow-versus-cycle confusion appearing early.

- 9** (a) carbon dioxide, methane or nitrous oxide **[1]** [2]
 (b) a matching activity: burning fossil fuels or deforestation (carbon dioxide); cattle farming, rice paddies or landfill (methane); nitrogen fertilisers or vehicle exhausts (nitrous oxide) **[1]**
The two marks must match each other. Naming methane and then writing "burning fossil fuels" scores the first only.
- 10** (a) **limewater**; it turns **milky / cloudy** **[1]** [2]
 (b) **expired** air, because it contains about a hundred times more carbon dioxide (about 4% against 0.04%) **[1]**
Both halves are needed in (a). "It goes white" is acceptable; "it changes colour" is not.
- 11** Any two of: expired air contains **more carbon dioxide**; contains **more water vapour** (saturated); is **warmer** **[2]** [2]
Temperature is the one most candidates miss, because they are looking for a gas. Nitrogen is unchanged and scores nothing.
- 12** the diaphragm contracts and flattens; the external intercostal muscles contract, pulling the ribs up and out **[1]** [3]
 the volume of the thorax increases **[1]**
 so the pressure inside the lungs falls below atmospheric pressure, and air flows in down the pressure gradient **[1]**
Order is the mark scheme. "The lungs suck air in" scores 0 for the whole question — nothing sucks, and a candidate who writes it has not understood any of the three steps.
- 13** very large total surface area (about 50 m² from 300 million alveoli) — more area for diffusion **[1]** [3]
 wall only one cell thick — a very short diffusion distance **[1]**
 dense capillary network with blood constantly flowing, and a moist lining — maintains the concentration gradient and lets gases dissolve before they diffuse **[1]**
Each mark needs feature plus consequence. Three bare features score 1 at most, and telling a candidate that once is worth more than marking it three times.
- 14** both ventricles pump the same volume of blood per beat **[1]** [3]
 the left ventricle pumps blood all round the body, a far greater distance against far greater resistance, while the right pumps only to the lungs **[1]**
 so the left ventricle must generate a much higher pressure, which needs more muscle **[1]**
"It holds more blood" scores 0 and contradicts the first marking point. Candidates who open with the same-volume sentence almost never then write the wrong reason.
- 15** artery wall thick with muscle and elastic tissue; vein wall thin **[1]** [3]
 artery lumen narrow; vein lumen wide **[1]**
 artery has no valves; vein has valves along its length **[1]**
The question said structural, so "arteries carry blood away from the heart" earns nothing here even though it is true. Marks are for the three physical differences.
- 16** the capillaries together have a far greater total cross-sectional area than the aorta (about 1000 times) **[1]** [3]
 the same volume of blood must pass every point each second, so a greater area means a lower speed **[1]**
 the slow flow gives time for diffusion, so exchange of oxygen, glucose and waste can be completed **[1]**
This is the hardest of the Section 2 questions. A candidate who says "capillaries are narrow so blood is slow" has the right answer for the wrong reason and scores the third mark only.

- 17** vena cava → right atrium → right ventricle **[1]** [3]
 → pulmonary artery → lungs → pulmonary vein **[1]**
 → left atrium → left ventricle → aorta **[1]**
Any route that misses the lungs, or that goes from the lungs straight to the aorta, loses everything after the error. The double circulation is the point of the question.
- 18** energy enters as light, is captured by producers and passes up the trophic levels in one direction **[1]** [3]
 it is lost as heat at every level and is never reused, which is why the Sun must keep supplying more **[1]**
 nutrients (carbon, nitrogen, mineral ions) are taken up, passed on, released by decomposers and taken up again — they cycle indefinitely **[1]**
"Energy is recycled" anywhere in the answer caps it at 1. This single sentence appears in most papers on this unit and is the most useful thing to correct.
- 19** only about 10% of the energy at one level is transferred to the next **[1]** [3]
 about 90% is lost as heat in respiration, in faeces and urine, and in parts that are not eaten **[1]**
 after four or five levels there is too little energy left to support another population **[1]**
"Energy is lost" without the figure scores 1. The 10% is what turns a vague answer into a full one.
- 20** place the quadrat at **random** positions, e.g. using random number coordinates **[1]** [3]
 count the dandelions inside it and **repeat** many times, then calculate the mean per quadrat **[1]**
 divide by the area of the quadrat to get a density per m², then multiply by the total area of the field **[1]**
The two words that carry marks are random and repeat. "Throw the quadrat over your shoulder" is accepted as random in most schemes but is worth discouraging.
- 21** fewer trees photosynthesising, so less carbon dioxide is removed from the atmosphere **[1]** [3]
 carbon stored in the wood is released as carbon dioxide when it is burned or decays **[1]**
 plus any one other: habitats destroyed and biodiversity lost; soil erosion and leaching once the roots are gone; less transpiration so reduced local rainfall **[1]**
Most candidates give the first mechanism only and then a consequence, scoring 2 of 3. The release from the felled wood is the missing mark almost every time.
- 22** short-wave radiation from the Sun passes through the atmosphere and warms the Earth's surface **[1]** [3]
 the warmed surface radiates energy back out as longer-wave infrared **[1]**
 greenhouse gases absorb this infrared and re-radiate it in all directions, including back towards the surface, so less heat escapes to space **[1]**
Any mention of the ozone layer or a "hole" is a different topic and scores nothing. It is worth naming that explicitly when returning the paper.
- 23** (a) $14 \times 480 = 6720 \text{ cm}^3$ per minute **[1]** = **6.72 dm³ per minute** **[1]** [5]
 (b) $68 \times 72 = 4896 \text{ cm}^3$ per minute **[1]** = **4.90 dm³ per minute** (4.896) **[1]**
 (c) her muscles respire faster, so they need more oxygen and produce more carbon dioxide; more air must be moved and more blood delivered each minute **[1]**
The conversion is where marks go: dividing by 100 instead of 1000 gives 67.2 and 48.96, which look wrong by inspection — a resting person does not breathe 67 dm³ a minute. Teach the sanity check, not the rule.

- 24** (a) $3.0 \times 10^8 \times 2.4 \times 10^{-7}$: multiply $3.0 \times 2.4 = 7.2$, add the powers $8 + (-7) = 1$ **[1]** = $7.2 \times 10^1 = 72 \text{ m}^2$ **[1]**
 (b) $9.0 \times 10^7 \times 4.0 \times 10^{-7} = 3.6 \times 10^1 = 36 \text{ m}^2$ **[1]**; reduction = $(72 - 36) \div 72 \times 100 = 50\%$ **[1]**
 (c) with half the surface area oxygen diffuses into the blood far more slowly; climbing stairs makes his muscles demand much more oxygen, which his lungs cannot supply, so he is breathless **[1]**
Note the trap in (b): his individual alveoli are larger (4.0 against $2.4 \times 10^{-7} \text{ m}^2$) and his total area is still halved, because there are so many fewer of them. A candidate who answers "his surface area went up" has reasoned from the wrong number and should redo it, not be told the answer.
- 25** (a) the **grasshopper** and the **rabbit** **[1]** [5]
 (b)(i) the grass **increases**, because fewer herbivores are eating it **[1]**
 (b)(ii) the fox population **falls at first but survives**, because it can eat more shrews instead **[1]**
 (c) most organisms in a web have more than one food source, so they can switch when one prey species is lost **[1]**; energy has alternative routes through the web, so losing one species does not break every link and the community stays stable **[1]**
In (b)(ii) an answer of "the fox dies out" scores 0 and shows the candidate read the web as two separate chains. Point at the shrew → fox arrow rather than explaining it.
- 26** (a) $48\,000 \times 0.1 = 4800 \text{ kJ/m}^2\text{/year}$ **[1]** [5]
 (b) $4800 \times 0.1 = 480$, then $480 \times 0.1 = 48 \text{ kJ/m}^2\text{/year}$ **[1]**; $(48 \div 48\,000) \times 100 = 0.1\%$ **[1]**
 (c) feeding small fish to larger fish adds an extra trophic level **[1]**; about 90% of the energy is lost at that transfer, so only about a tenth as much energy reaches people as would if they ate the small fish directly **[1]**
The level-counting error gives 480 in (b) instead of 48. Ask which organism is the tertiary consumer before correcting the arithmetic — the arithmetic is usually right, the counting is not.
- 27** (a) **300 m** **[1]** [5]
 (b) decrease = $48 - 11 = 37 \text{ mg/dm}^3$ **[1]**; $(37 \div 48) \times 100 = 77.1\%$ **[1]**
 (c) the nitrate causes algae to grow rapidly (an algal bloom), which blocks light so the plants on the river bed die **[1]**; decomposer bacteria multiply on the dead material and use up the dissolved oxygen in their aerobic respiration **[1]**
"The nitrate uses up the oxygen" scores 0 in (c) — nitrate does nothing to oxygen directly. The chain runs nitrate → algae → light → dead plants → bacteria → oxygen, and the two most-dropped words are light and aerobic.

Biology Mock 2 — mark scheme

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- 1** (a) rings of **cartilage** **[1]** [2]
 (b) they hold the trachea open / stop it collapsing when air is drawn in at low pressure **[1]**
"To protect it" scores 0. The rings solve a specific mechanical problem, and naming that problem is the mark.
- 2** (a) **goblet cells**; the mucus traps dust, pollen, bacteria and other particles **[1]** [2]
 (b) cilia beat to sweep the mucus and trapped particles up the trachea to the throat, where it is swallowed **[1]**
Both halves needed in (a). A candidate who has these two cells the wrong way round will also get the smoking question wrong — check Q13 before deciding it was a slip.

- 3** (a) glucose + oxygen → carbon dioxide + water (energy released) **[1]** [2]
 (b) in the **mitochondria** **[1]**
“Energy is made” or “energy is produced” loses the mark in some schemes. Energy is released from glucose; it is not created.
- 4** (a) the **pulmonary artery** **[1]** [2]
 (b) the **right ventricle** **[1]**
“Right atrium” in (b) is common. Blood always leaves the heart from a ventricle; atria only pass it downwards.
- 5** Any two of: **biconcave disc shape** — large surface area and a short diffusion distance to the centre; **no nucleus** — more room for haemoglobin; **contains haemoglobin** — binds oxygen to form oxyhaemoglobin **[2]** [2]
A shape with no consequence scores half an idea and no mark. “It is small so it fits through capillaries” is true but is not an adaptation for carrying oxygen, which is what the question asked.
- 6** (a) **population**: all the members of one species living in an area **[1]** [2]
 (b) **ecosystem**: all the organisms in an area together with the non-living parts of their surroundings, interacting **[1]**
An ecosystem definition with no mention of non-living factors scores 0. This is a one-word fix and worth insisting on.
- 7** (a) the direction in which **energy** (and biomass) is transferred, from the organism eaten to the organism eating it **[1]** [2]
 (b) a producer makes its own food by **photosynthesis**, capturing light energy **[1]**
“Shows what eats what” scores 0 — it describes the diagram without saying what the arrow means.
- 8** (a) about **10%** **[1]** [2]
 (b) lost as **heat** released during respiration **[1]**
Accept faeces/urine or uneaten parts in (b), but heat is the largest route and the one to teach.
- 9** (a) **leaching** (accept run-off) **[1]** [2]
 (b) **algae** (an algal bloom) **[1]**
“Bacteria” in (b) is the right organism at the wrong step — they come in later, feeding on the dead material.
- 10** Any two of: deforestation and habitat destruction; pollution (fertilisers, sewage, pesticides, plastics); overfishing or over-hunting; climate change from burning fossil fuels **[2]** [2]
Both marks need distinct activities. “Pollution” twice, in two forms, is acceptable only if the two forms are clearly different.
- 11** (a) breathing (ventilation) is the muscular movement of air in and out of the lungs; respiration is a chemical reaction that releases energy from glucose **[1]** [2]
 (b) in **every living cell** of the body (in the mitochondria) **[1]**
“Respiration takes place in the lungs” in (b) scores 0 and is the single most repeated error in the whole of Unit 2.
- 12** the diaphragm **relaxes** and moves up into its domed shape; the intercostal muscles relax so the ribs move down and in **[1]** [3]
 the volume of the thorax decreases **[1]**
 so the pressure inside the lungs rises above atmospheric pressure and air flows out **[1]**
“The diaphragm pushes the air out” scores 0 — it describes a relaxed muscle doing work it cannot do. At rest, exhalation is elastic recoil.

- 13** tar paralyses and destroys the cilia, so mucus is not swept out; it accumulates, causing coughing and repeated infection **[1]** [3]
 alveolar walls break down and alveoli merge into fewer, larger sacs (emphysema), reducing the surface area for gas exchange **[1]**
 carbon monoxide binds to haemoglobin in place of oxygen, so the blood carries less oxygen **[1]**
The third point is about transport, not gas exchange, and is the one that separates a 3 from a 2. Marks are for three different mechanisms, not three symptoms.
- 14** blood in a vein is at very low pressure and moves slowly, so it could flow backwards **[1]** [3]
 valves close to prevent backflow, so blood travels only towards the heart (helped by skeletal muscles squeezing the veins) **[1]**
 blood in an artery is at high pressure straight from the ventricle, so backflow cannot occur and valves are unnecessary **[1]**
Most answers cover the veins and stop. The question has two halves and the artery half is a full mark.
- 15** blood passes through the heart **twice** for each complete circuit of the body **[1]** [3]
 the pulmonary circuit runs heart → lungs → heart; the systemic circuit runs heart → body → heart **[1]**
 advantage: blood is pumped again after the lungs, so it travels to the body at high pressure and delivers oxygen quickly, supporting a high rate of respiration and activity **[1]**
A comparison with a fish's single circulation is a legitimate route to the third mark and is worth praising.
- 16** wall one cell thick — a very short diffusion distance between blood and cells **[1]** [3]
 very narrow lumen, roughly one red blood cell wide — slows the blood and brings it close to the wall **[1]**
 enormous number of them, forming a dense network with a huge total surface area — more area for exchange and no cell far from a capillary **[1]**
Feature plus consequence on every line. Three bare features score 1.
- 17** pressure falls continuously from the aorta (about 16 kPa) to the vena cava (about 0.5 kPa) **[1]** [3]
 most of the fall happens across the arterioles and capillaries **[1]**
 blood returns because skeletal muscles contract and squeeze the veins, and valves ensure the squeeze moves blood only towards the heart **[1]**
"The heart sucks it back" scores 0. If a candidate writes it, the muscle-pump idea has not landed at all.
- 18** a pyramid of numbers counts organisms and ignores their size **[1]** [3]
 so one very large producer, such as a single tree supporting hundreds of caterpillars, gives a narrow bottom bar and an inverted pyramid **[1]**
 a pyramid of biomass measures the total mass at each level, and the tree's mass is far greater than all the caterpillars together, so it comes out the usual way up **[1]**
A worked example (one oak, 500 caterpillars) is the clearest route to all three marks and should be encouraged over abstract phrasing.
- 19** decomposers (bacteria and fungi) feed on dead organisms, dead plant material and waste, breaking them down **[1]** [3]
 this releases mineral ions such as nitrates back into the soil, where plants absorb them again **[1]**
 without them, dead material would accumulate and the mineral ions locked inside it would not be returned, so plant growth would eventually fail **[1]**
"Everything would smell" is not the answer. The third mark is about nutrients becoming unavailable, not about tidiness.

- 20** a line (tape or rope) is laid across the area being studied **[1]** [3]
 quadrats are placed at fixed, regular intervals along the line and the organisms in each are counted or their percentage cover recorded **[1]**
 a transect is used when studying how a population changes along a gradient — up a shore, out from a path, into shade — where random placement would hide the very pattern being investigated **[1]**
The third mark is the one that matters. A candidate who describes the method but cannot say when it beats random sampling has learned a procedure, not a technique.
- 21** extra greenhouse gases trap more of the infrared radiated by the Earth, so the mean global temperature rises **[1]** [3]
 ice sheets and glaciers melt, and seawater expands as it warms, so sea level rises **[1]**
 low-lying habitats are flooded and lost; species that cannot move or adapt fast enough die out **[1]**
Thermal expansion is the half most candidates miss — they name melting ice only. Both contribute, and expansion is roughly comparable in size.
- 22** use less fertiliser, and do not apply it just before heavy rain **[1]** [3]
 leave an unfertilised buffer strip of vegetation along the riverbank to catch run-off **[1]**
 treat sewage before discharging it, so it does not add nitrates and organic matter directly **[1]**
“Stop farming” scores 0. The question asks for measures, which means things a farmer could actually do next season.
- 23** (a) rest: $15 \times 500 = 7500 \text{ cm}^3 = \mathbf{7.5 \text{ dm}^3 \text{ per minute [1]}}$; exercise: $28 \times 1750 = 49\,000 \text{ cm}^3 = \mathbf{49 \text{ dm}^3 \text{ per minute [1]}}$ [5]
 (b) $49 \div 7.5 = \mathbf{6.5 \text{ times greater (6.53) [1]}}$
 (c) the muscles respire faster, so they need more oxygen and produce more carbon dioxide **[1]**; receptors detect the rising carbon dioxide and the brain increases both the rate and the depth of breathing, so more air is exchanged each minute **[1]**
In (c), “because he needs more oxygen” alone scores 1. The second mark is for the carbon dioxide being the signal — which is also the answer to why holding your breath becomes unbearable.
- 24** (a) 85% still works: $200 \times 0.85 = \mathbf{170 \text{ cm}^3 \text{ per dm}^3 [1]}$ [5]
 (b) $5.2 \times 170 = \mathbf{884 \text{ cm}^3 \text{ per minute [1]}}$; non-smoker $5.2 \times 200 = 1040$, so **156 cm³ per minute less [1]**
 (c) the limit is **transport**, not gas exchange **[1]**; oxygen diffuses into his blood normally, but 15% of his haemoglobin is permanently occupied, so every dm³ leaving his lungs is 15% short; walking uphill raises his muscles’ demand and he cannot meet it **[1]**
 $200 - 15 = 185$ in (a) is the error this question exists to catch — a percentage of a quantity is not a subtraction. Everything downstream then follows from a wrong number, so award follow-through in (b).
- 25** (a) total = $3+5+2+6+4+3+5+4 = 32$, mean = $32 \div 8 = \mathbf{4 \text{ per quadrat [1]}}$; quadrat area = $0.25 \times 0.25 = 0.0625 \text{ m}^2$, density = $4 \div 0.0625 = 64 \text{ per m}^2$, total = $64 \times 250 = \mathbf{16\,000 \text{ plantains [1]}}$ [5]
 (b) take many more quadrats, and place them using random number coordinates rather than by eye **[1]**
 (c) predator numbers can only rise once prey is already plentiful, and breeding takes time, so the predator increase lags behind **[1]**; the increased predation then reduces the prey, and the predators decline afterwards through lack of food **[1]**
The quadrat area is the trap: 0.25 m on a side is 0.0625 m², not 0.25 m². Using 0.25 gives 4000, which is a quarter of the truth and looks entirely plausible. Award the mean mark and withhold the second.

- 26** (a) $480\,000 \times 5.5 = 2\,640\,000$ tonnes **[1]** = **2.64×10^6 tonnes per year** **[1]** [5]
(b) $12\,000 \times 5.5 =$ **66 000 tonnes per year** of absorption lost **[1]**
(c) the carbon stored in the felled trees is also released as carbon dioxide when the timber is burned or decays **[1]**; and the loss is cumulative — another 12 000 hectares goes every year, so the absorption lost grows year on year (accept: carbon is also released from the disturbed soil) **[1]**

Part (c) separates the strong candidates. Most give one reason; the cumulative point and the released-carbon point are genuinely different and both are available.

- 27** (a) the definition is about **direction**, not oxygen: arteries carry blood away from the heart and veins carry it towards the heart **[1]**; the pulmonary artery carries deoxygenated blood and the pulmonary vein carries oxygenated blood **[1]** [5]
(b) nothing in the chain is a poison — nitrate is a plant nutrient **[1]**; it causes an algal bloom that blocks the light, the plants below die, decomposer bacteria multiply and use up the dissolved oxygen in aerobic respiration, and the fish suffocate **[1]**
(c) energy is **not** recycled — it flows through and is lost as heat; it is **nutrients** that decomposers recycle **[1]**

Marking someone else's error is the same cognitive work as marking your own and far easier to face. If a candidate corrects all three here but made the same mistakes earlier in the paper, that is worth showing them side by side — it is usually the most useful two minutes of the whole review.